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Responsive Architecture in Practice: BIM/DT/AI/IoT for Dynamic Fire Evacuation—A Comparative Case Study Analysis

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Abstract

This study presents a comparative analysis of six DFS implementations representing different maturity levels and investigates the systemic gap between technological capabilities and regulatory approaches. A structured narrative review with case-based analysis was conducted using the Scopus database (2015–2026) with six targeted queries. The case selection followed the PICo protocol. An original ten-criterion DFS maturity assessment rubric—grounded in the Technology Readiness Level (TRL), Integration Readiness Level (IRL), and Digital Twin Maturity Model frameworks—was applied to all six cases. Inter-rater validation yielded substantial agreement ($\kappa_W = 0.797$; unweighted $\kappa = 0.674$ [95% CI: 0.509, 0.839]). The results indicate a clear maturity gradient (Dimension X: 4–9 points; Dimension Y: 2–8 points). Benefits reported in the analysed primary studies include up to a 55 s reduction in evacuation time, a 72% improvement compared with static signage, and a 34-percentage-point increase in evacuation success rate under simulation-based conditions. Five normative recommendations are proposed to address the structural regulatory gap between current prescriptive frameworks and DFS deployment in Poland and the EU. This study argues that prescriptive rules should remain the baseline, whereas complex facilities may adopt performance-based DFS solutions, provided that equivalence to conventional protection levels is rigorously demonstrated. From a sustainability perspective, the study frames DFS as a dynamic safety layer that supports occupant protection, operational resilience, and lifecycle adaptability in complex buildings exposed to uncertain fire and crowd conditions.

Keywords: sustainability; fire safety engineering; dynamic fire safety systems (DFSs); performance-based design; smart buildings; digital twin; building information modelling (BIM); internet of things (IoT); artificial intelligence (AI); cyber–physical systems; evacuation modelling; resilience; risk management; safety engineering; regulatory frameworks

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1. Introduction

1.1. Digital Transformation of Fire Safety

Contemporary buildings function as complex technical systems in which fire alarm systems (FASs), voice alarm systems (VASs), building management systems (BMSs), smoke

extraction ventilation, and video surveillance are expected to operate as a single coherent entity during an emergency. The traditional RBS paradigm was founded on static fire scenarios developed in the design stage—documents that are rarely revised after the building is handed over—while the actual behaviour of the building during a fire incident progressively diverges from the original design assumptions.

Building information modelling (BIM) has become an established design standard; however, once a building is handed over, the BIM model is typically no longer maintained. This is particularly important because BIM was originally conceived as a lifecycle-oriented approach that can support planning, design, construction, and facility operations within a single digital model [1]. This limitation is consistent with the broader literature, which shows that the implementation of BIM in existing buildings remains constrained by the high effort required for model creation, the difficulty of updating information, and the uncertainty of available building data [2]. The proliferation of Internet of Things devices and continuous operational data streams has created the conditions necessary for the concept of a living Digital Twin (DT)—a model continuously fed by sensor data. Integration of BIM with IoT technologies enables real-time data exchange between a physical asset and its digital representation, forming the technological foundation of Digital Twin systems [3]. This trend is consistent with recent review research showing that IoT-enabled fire safety systems increasingly integrate detection, localisation, and evacuation functions through sensor networks, real-time data handling, and advanced analytics [4]. Artificial intelligence adds predictive capability and dynamic evacuation routing. Recent studies have demonstrated that intelligent evacuation guidance systems can dynamically generate safer and faster routes in large buildings by combining hazard intensity, congestion data, and distributed intelligent indicators [5]. This observation is consistent with recent research showing that digital twins can support building safety throughout the lifecycle by integrating sensors, 3D visualisation, and data analytics for threat detection and monitoring [6]. Together, these technologies constitute a responsive architecture: a building that actively responds to hazards rather than passively awaiting the activation of predefined static scenarios. This fragmentation of systems becomes particularly problematic under emergency conditions, where coordinated and timely decision-making is critical to ensure occupant safety.

The tension between technological capability and regulatory constraints constitutes a critical and insufficiently addressed gap in contemporary fire safety engineering, particularly in the context of integrating real-time data-driven evacuation systems into existing prescriptive regulatory frameworks. This gap is particularly consequential in complex and high-occupancy buildings, where static evacuation assumptions may not reflect dynamically evolving fire conditions and occupant behaviour.

The sustainability dimension of this problem should not be reduced to environmental indicators alone. In complex buildings, sustainable performance also includes the capacity of the built environment to maintain safe, reliable, and adaptable operation throughout its life cycle. From this perspective, dynamic fire safety systems contribute to sustainability by transforming static fire safety provisions into an operational layer capable of responding to changing hazard development, occupant distribution, and system conditions. Such an approach links human-centred safety, resilience, and adaptive facility management within a performance-oriented framework for sustainable building operation.

1.2. Objective, Scope, and Research Questions

Despite rapid technological advances, there is a lack of structured comparative analyses that link the maturity, operational performance, and regulatory applicability of DFS solutions. This study aims to demonstrate that DFS solutions are already being tested or deployed in existing buildings and built infrastructure and to identify the structural

regulatory discrepancies that impede their wider adoption. Three research questions were formulated: (RQ1) How does the sensor → model → decision → actuator chain function in the six cases analysed? (RQ2) Which technological components are essential for a functioning DFS? (RQ3) How can DFS solutions be transposed into the Polish legal system? The original contribution of this study is the development of a structured DFS maturity assessment rubric and its application to real-world case studies, enabling a systematic comparison between technological integration and operational effectiveness in the context of regulatory constraints. By doing so, the study also positions DFS maturity assessment as a tool for evaluating how digital fire safety systems may contribute to sustainability-oriented building performance through resilience, adaptive operation, and improved life safety reliability.

2. Review of the Literature

2.1. The DFS Concept

The DFS constitutes a closed-loop cyber–physical architecture composed of six interconnected layers [7], as illustrated in Figure 1.

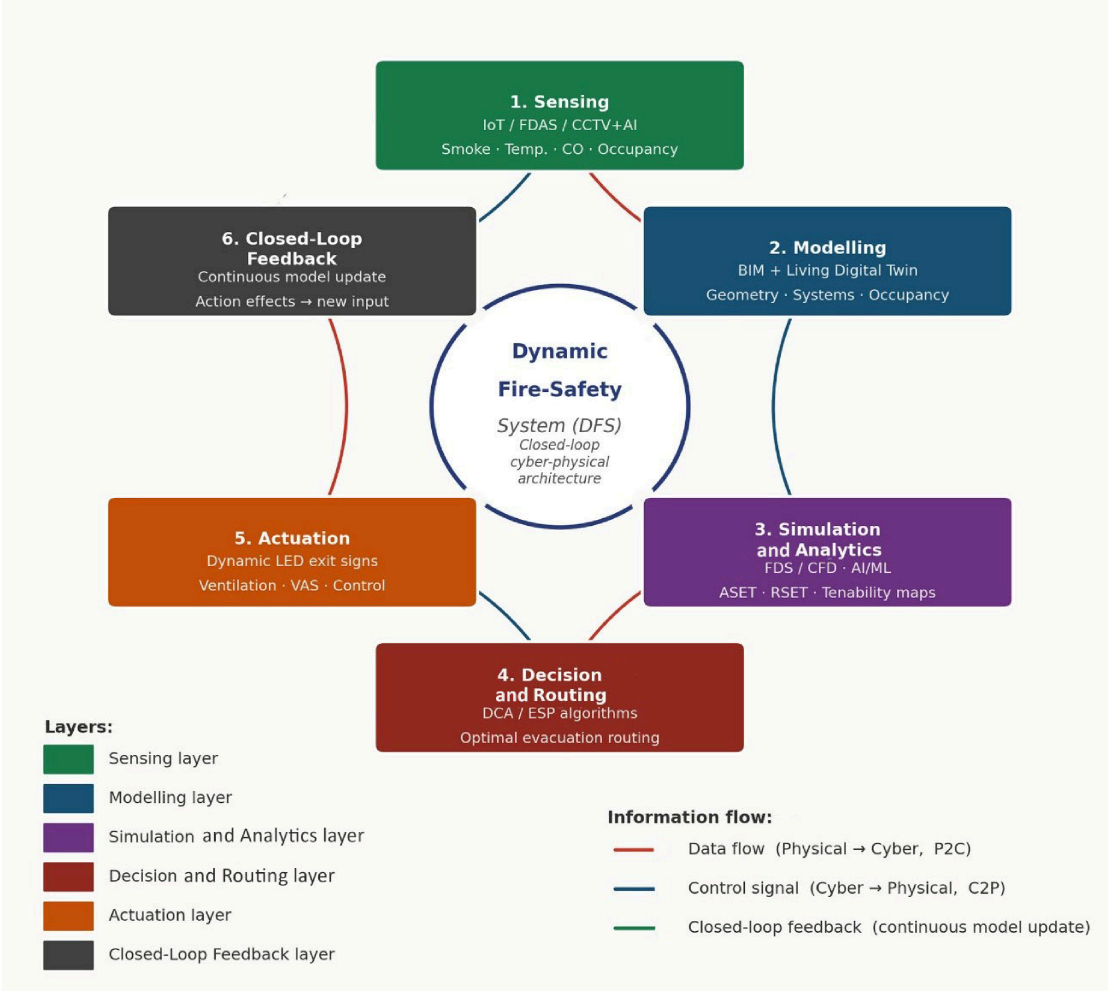


Figure 1. Closed-loop architecture of the dynamic fire safety system (DFS)—six layers from sensing to feedback. Authors' own elaboration.

The six layers form a continuous loop: Sensor data feed the living DT, which dynamically computes the available safe exit time (ASET), required safe exit time (RSET), and congestion risk; decision algorithms identify safe evacuation routes; dynamic exit signage

physically executes decisions; and the effects of each action are fed back into the model as new input data. This architecture is fundamentally distinct from the RBS, in which the decision chain is one-off and open.

2.2. BIM and Digital Twins in Fire Safety Engineering

Recent studies indicate a clear transition in fire safety engineering from static, document-based approaches to integrated, data-driven decision environments. Almatared et al. [7] emphasised that conventional fire safety management still suffers from fragmented information, limited interoperability, and insufficient real-time support, while digital twin-based frameworks can improve coordination between BIM, IoT, AI, and facility management processes. Lin et al. [8] further show that the evolution of emergency evacuation research increasingly relies on digital twins that integrate buildings, occupants, sensing, simulation, and decision-making within a continuous feedback loop. Furthermore, Zhang et al. [9] demonstrated that AIoT-enabled digital twin systems can extend this concept to operational fire safety management by supporting real-time fire recognition, prediction, visualisation, and emergency response.

Against this background, Tables 1 and 2 synthesise the main differences between traditional Rule-Based Safety (RBS) and dynamic fire safety systems (DFSs), as well as between successive digital modelling approaches, from static BIM to living Digital Twins. The comparison highlights a consistent pattern reported in the literature: as integration, real-time data exchange, and adaptive control increase, the practical relevance of the model for evacuation support and fire safety management increases. This distinction is consistent with studies showing that dynamic personnel guidance can improve evacuation efficiency compared to static signage [10].

Within this framework, real-time multimodal sensor fusion can be understood as an enabling mechanism of the six-layer DFS architecture because it supports the integration of thermal, optical, gas, and occupancy data streams within a unified digital model; moreover, it may improve fire localisation accuracy, situational awareness, and evacuation decision latency.

Table 1. Comparison of digital tools in fire safety engineering.

Feature	RBS (Rule-Based Safety)	DFS (Dynamic Fire Safety System)	Comment
Decision logic	Static, defined once at the design stage	Adaptive, responding to real-time sensor data	DFS is dynamically adjusted during an incident
Evacuation routing	Nearest-exit principle; fixed routes	Routes updated in real time based on tenability and congestion	DFS avoids routes that become unsafe
Scenario update	One-off (design stage); difficult to modify in operation	Continuous during an incident; automated	DFS responds to unforeseen events
Building model	Static paper documentation or outdated BIM	Living Digital Twin is continuously updated with sensor data	DT reflects the actual state of the building
Integration of systems	FAS, VAS, and BMS operate independently	Integrated cyber–physical ecosystem	Elimination of information silos
Evacuation effectiveness	Baseline (normative compliance)	Demonstrably superior in the cases analysed	Depends on the degree of DFS integration
Regulatory status	Full compliance with prescriptive requirements	No formal acceptance pathway in Poland or the EU	Primary barrier to DFS deployment

Source: Authors' own elaboration based on [1–3]. Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

Table 2. Comparison of digital modelling approaches (static BIM → living Digital Twin).

Feature	Static BIM	FDS Simulations	Agent-Based Models	Living Digital Twin (DFS)	Relevance to DFS
Model currency	Static (manually updated)	One-off (design stage)	Typically static	Continuously updated with sensor data	Greatest DT advantage
Real-time integration	None	None	Limited	Full, bidirectional (IoT + BMS)	Enable closed-loop functionality
Operational timing	Design phase only	Pre-incident	Pre-incident	During the incident (near-real-time)	Critical for evacuation
Prediction scope	Geometry and materials	Fire development—1 scenario	Occupant behaviour	Fire + occupants + building systems	Broadest scope
Adaptability	None	None	Limited	Full adaptation during the incident	Core of DFS
Validation level	High (geometry)	Moderate (experiments)	Moderate	Low/moderate (simulations and lab)	Greatest DT gap

Source: Authors' own elaboration; data compiled by the authors from the cited sources [1–3]. Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

2.3. Paradigm Shift: RBS → PBD → DFS

Meacham [11–13] identified the asymmetry of expertise between FSE practitioners and regulatory bodies as the primary institutional barrier to performance-based design (PBD): engineers possess considerably greater technical knowledge than the regulators responsible for approving their submissions. Chow et al. [14], employing the Cynefin framework, demonstrate that prescriptive requirements are applied mechanically, without a critical assessment of their suitability to specific building configurations. This critique is consistent with recent simulation-based research showing that an overreliance on prescriptive evacuation definitions can reduce the effectiveness of the design and lead to exit capacity provisions that are more conservative than necessary under comparable conditions [15]. Węgrzyński and Sulik [16], writing in the Polish academic context, document how the existing legal framework itself impedes the diffusion of FSE advances into professional practice. A similar relationship between regulatory evolution and the development of fire engineering practice has also been documented in other national contexts, including China [17]. Abualdenien et al. [18] note that FSE was one of the disciplines that remained long outside the integrated digital approaches, despite building fire safety being fundamental to building function. This larger shift is also reflected in recent bibliographic research showing that machine learning in fire safety is increasingly applied to evacuation, early warning, fire detection, and predictive risk assessment in the built environment [19].

3. Research Methodology

3.1. Study Design and the PICo Framework

This study employs a structured narrative review with a case-based analysis methodology. The searches were conducted in Scopus using six targeted query strings (2015–2026). The selection of cases followed the PICo framework: Population—buildings of occupancy categories ZL I–III (Polish classification) or critical infrastructure; Interest—DFS implementations that incorporate at least two components of the set BIM/DT/IoT/AI/dynamic signage; *Context*—the structural tension between the RBS and PBD/DFS paradigms; and *Outcomes*—a systemic gap matrix, evidence of DFS effectiveness, and normative recommendations. To ensure reproducibility, the six Scopus query strings used in the literature search are listed in

Table A1 (Appendix A). The total number of records retrieved was approximately 81 unique records (after deduplication of~147 raw results in all six strings); 27 met the PICo inclusion criteria in the title/abstract screening; six case studies were selected for full-text analysis based on the presence of at least two components of DFS, the availability of quantitative performance data, and peer-reviewed publication status. This selection process is summarised in Figure A1 (Appendix A). The methodology adopted in this study is illustrated in Figure 2. The structured narrative review methodology was selected over a full systematic review Torgerson [20] on the basis of methodological fit: the study objective is conceptual framework development and comparative case analysis rather than effect estimation in a homogeneous pool of interventions. Accordingly, the objective of the search strategy was not to identify and quantitatively synthesise all publications on BIM, Digital Twins, IoT, AI, or evacuation modelling in fire safety. Instead, the searches were designed to identify a focused set of information-rich, peer-reviewed implementations in which at least two DFS components were sufficiently documented to allow reconstruction of the sensor–model–decision–actuation chain. The six included cases should therefore be interpreted as analytically selected exemplars representing different maturity levels of DFS development, rather than as a statistically representative sample of the entire research field. This approach is consistent with established practice in fire safety engineering, where frameworks are developed through structured case comparison rather than meta-analytic aggregation Almatared et al. [7]; Chow et al. [14]; Nasery et al. [21]; Wang et al. [22].

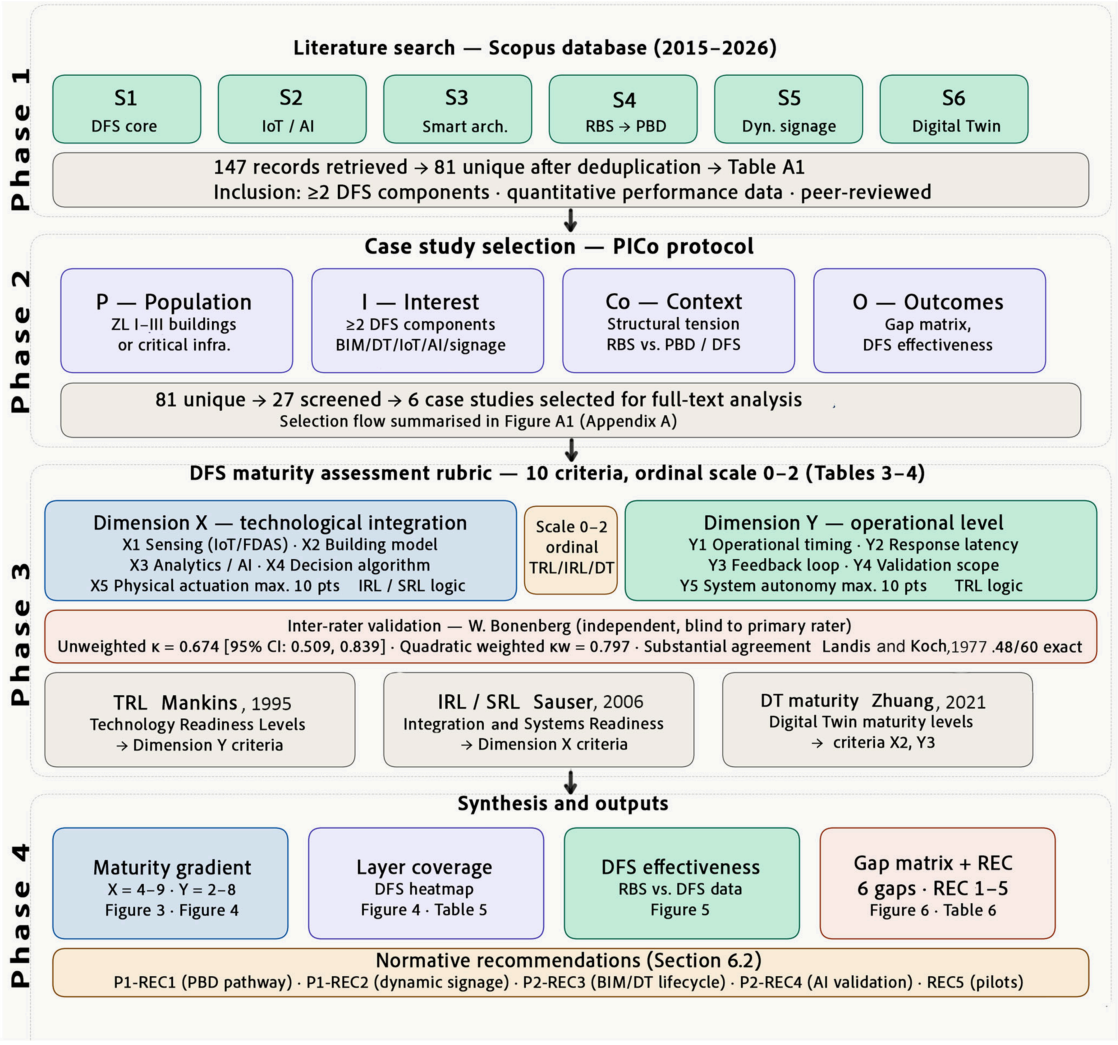


Figure 2. Overview of research methodology: four-phase structured narrative review with case-based analysis. Phase 1: Scopus literature search (six query strings S1–S6; Table A1). Phase 2: PICo case

selection protocol (81 unique records → 27 selected → 6 case studies; Figure A1). Phase 3: DFS maturity assessment rubric (10 criteria, ordinal scale 0–2; inter-rater validation $\kappa = 0.674$, $\kappa w = 0.797$). Phase 4: synthesis of results and normative recommendations. Authors' own elaboration.

3.2. DFS Maturity Assessment Rubric—Scoring Criteria and Methodological Foundations

Purpose and general structure: The DFS maturity assessment rubric (Table 3) was developed as an original instrument by synthesising the established frameworks of Technology Readiness Levels (TRLs), Integration Readiness Levels (IRLs), Systems Readiness Levels (SRLs), and the Digital Twin Maturity Model. Each scoring threshold (0 = absent, 1 = partial, 2 = full/mature) was operationalised with explicit definitions directly linked to the implications of fire safety performance. For example, a score of 2 in X5 (physical actuation) requires automated, immediate control of dynamic exit signs without operator intervention—a feature that directly reduces Required Safe Egress Time (RSET) and prevents occupants from being directed toward routes that have lost tenability. Similarly, a score of 2 in Y2 (response latency) requires a system reaction time of $\leq$10 s, which is consistent with the ASET margins documented in the analysed case studies; a score of 2 in Y4 (validation scope) corresponds to full-scale physical validation, reflecting the TRL 7–8 threshold for regulatory acceptance of non-standard fire safety solutions. The rubric was developed, applied, and interpreted by Przemysław Konopski. This limitation is methodologically significant and is discussed in Section 3.3.

The methodological basis of the rubric combines Technology Readiness Level logic [23,24], Integration and System Readiness Level concepts [25], and Digital Twin maturity modelling [26].

The full scoring matrix is presented in Table 3. Each criterion has explicitly defined thresholds for each of the three possible scores (0, 1, 2). Table 3 provides complete operationalisations together with methodological justification that refers to TRL, IRL, or the Digital Twin Maturity Model. Dimension X captures the level of technological integration (sensing, modelling, analytics, decision-making, and actuation), while Dimension Y reflects operational performance (timing, latency, feedback, validation, and autonomy).

Each criterion is evaluated independently using explicitly defined thresholds corresponding to absent (0), partial (1), or full/mature (2) implementation. The physical meaning of each threshold is anchored to fire safety performance requirements: a score of 2 on X1 requires a sensor density sufficient to localise fire events within 10 s; a score of 2 on X5 requires automated actuation without operator intervention; a score of 2 on Y2 corresponds to a system reaction time of 10 s or less, which is consistent with ASET margins in the analysed cases; a score of 2 on Y4 requires full-scale physical validation, reflecting the TRL 7–8 threshold for regulatory acceptance of non-standard fire safety solutions.

Following joint discussion of all 12 discrepancies in the assessments, the primary rater's scores (P.K.) were retained in all cases, as the source descriptions did not provide sufficient evidence to confirm the higher score assigned by the second rater. The consensus discussion was conducted as an evidence-tracing procedure rather than as a numerical averaging exercise. For each of the 12 discrepant scoring cells, both raters returned to the relevant primary source and compared the reported technical description with the operational thresholds defined in Table 3. A higher score was retained only when the source explicitly documented the required functionality, validation level, response latency, or actuation mechanism. Where the available description was ambiguous or did not directly confirm the higher threshold, the more conservative score was maintained. This explains why all discrepant cases were ultimately resolved in favour of the primary rater: not because the second assessment was disregarded, but because the available source evidence did not justify increasing the score under the predefined rubric criteria. The

inter-rater study thus serves as a validation of the rubric instrument rather than a basis for score aggregation: the substantial agreement (κw = 0.797) indicates that the rubric criteria are sufficiently explicit to produce consistent scoring by independent trained raters. Full inter-rater data (Table A2), the confusion matrix (Table A3), and kappa computation details (Table A4) are provided in Appendix B.

The inter-rater reliability of the DFS maturity assessment rubric was tested by two independent experts (P.K. and W.B.) on all 60 scoring cells. Exact agreement was achieved in 48 cells (80.0%). These results confirm the acceptable to good reliability of the rubric, although some subjectivity remains in borderline cases, particularly regarding the degree of physical actuation (X5) and real-time analytics (X3). The obtained level of agreement supports the use of the rubric as a structured comparative tool, while highlighting the need for further refinement and broader multi-rater validation in future studies.

Dimension Y (Operational Level)—alignment with TRL: Criteria Y1–Y5 are conceptually aligned with the Technology Readiness Level (TRL) scale introduced by Mankins [23]. TRL is a nine-level ordinal scale ranging from TRL 1 (observation of basic principles) to TRL 9 (system proven in an operational environment). Héder [24] documents its evolution from a NASA instrument to a standard of the European Commission in Horizon Europe research programs. The central dimension of TRL progression is the validation environment and the degree of system autonomy, which is precisely what the criteria Y4 (validation scope) and Y5 (system autonomy) measure. TRL is not applied directly in this study; rather, its logic is adapted to the DFS domain: the three-level scale 0–1–2 replaces the nine-level TRL scale, and the criteria are tailored to fire safety system specifics.

Dimension X (Technological Integration)—alignment with IRL and SRL: Whereas TRL assesses the maturity of a single technology, the DFS is by definition a multi-component system. Sauser et al. [25] proposed the Integration Readiness Level (IRL), a scale that evaluates the degree to which two or more technologies have been successfully integrated into a functioning subsystem. The combination of TRL and IRL yields the system readiness level (SRL), defined as the normalised product of TRL and IRL. SRL measures the overall maturity of a composite system, accounting for both the maturity of the components and the quality of their mutual integration. Criteria X1–X5—covering sensing, building model, analytics/AI, decision algorithm, and physical actuation—operationalise the IRL logic: each criterion measures not the mere presence of a component, but the depth of its integration with the remaining DFS layers. A system with advanced IoT sensors (X1 = 2) but no physical actuation (X5 = 0), such as the Lille case, obtains a low total X = 4, which is methodologically consistent with IRL: technology without system integration does not constitute a mature solution.

Digital Twin Maturity Models: **Zhuang et al. [26] define five levels of DT maturity: (1) geometric, (2) informational, (3) integrated sensor-data, (4) predictive, and (5) autonomous closed-loop.** Criteria X2 (building model) and Y3 (feedback loop) directly correspond to this progression: X2 = 0 represents Zhuang's Levels 1–2; X2 = 2 combined with Y3 = 2 corresponds to Levels 4–5. Among the six cases analysed, only the study by Zhang et al. [9] reaches level 5, and only at the demonstration stage of development.

The reliability between raters was assessed by a co-author (W. Bonenberg), who independently applied the rubric to all six case studies based on condensed case descriptions, without prior knowledge of the primary rater's scores. Cohen's kappa was calculated in all 60 criterion–case assessments. The unweighted kappa was κ = 0.674 [95% CI: 0.509, 0.839], indicating substantial agreement. The quadratic weighted kappa, which is appropriate for ordinal scales, was κw = 0.797, also indicating substantial agreement.

Table 3. Full scoring matrix of the DFS maturity assessment rubric—definitions for the 0/1/2 scale.

Criterion	Definition	0 pts—Absent/Not Applicable	1 pt—Partial	2 pts—Full/Mature	TRL/IRL/DT Basis
DIMENSION X—Technological Integration (Criterion X1–X5)					
X1	Sensing (IoT/FDAS/CCTV)	No operational sensor data; building information limited to design stage documents (drawings, static calculations)	Sensors installed but operating unidirectionally (binary alarm) or infrequently; no continuous data stream	Dense IoT sensor network providing continuous data streams (temperature, smoke, CO, occupancy); data geolocated within building model	IRL P2C (physical-to-cyber) integration criterion
X2	Building model (BIM/DT)	No digital model or 2D CAD only; alternatively, a static BIM model unused in building operations	Static 3D BIM model with geometry and fire compartment boundaries; not updated with operational data	Living Digital Twin: 3D BIM model automatically updated with sensor data in real-time or near-real-time mode	DT Maturity Level 1 to 2 (static) vs. 3 to 5 (living); Zhuang et al. [26]
X3	Analytics/AI	No computational analysis or manual calculations only; alternatively, pre-computed scenarios without updates	One-off FDS or Pathfinder simulations at the design stage; the outputs are not updated during building operation	AI/ML algorithms or FDS operating in real-time or near-real-time mode; capable of adapting to the current hazard state	TRL prediction criterion: from one-off simulation to adaptive real-time analysis
X4	Decision algorithm	The Static Dijkstra algorithm or equivalent—identifies the geometrically shortest route without regard to current hazard	Algorithm accounting for hazard location and intensity (e.g., edge-weight modification in a graph); does not address congestion	Multi-criteria algorithm combining thermal and toxic hazards with congestion risk (e.g., DCA, ESP)—the route is optimised across multiple objectives	IRL integration criterion: sensing layer linked to decision layer
X5	Physical actuation	No physical update of signage or systems; system output accessible only through a digital interface (application, WebGL)	Partial control: manual sign switching by operator or control of a single subsystem only (e.g., ventilation only)	Automated, immediate control of dynamic LED exit signs and/or ventilation without operator intervention in the loop	IRL C2P (cyber-to-physical) loop closure criterion
DIMENSION Y—Operational Level (criterions Y1-Y5)					
Y1	Operational timing	The system operates exclusively at the design stage; it provides no operational decisions during the use of the building or incident	The system provides decision support to the operator during an incident (recommendations, visualisation); decisions require human approval	The system operates autonomously during a real or simulated incident without requiring operator confirmation	TRL: design stage (TRL 4) → operational support (TRL 6) → autonomous operation (TRL 7–8)

Table 3. *Cont.*

Criterion	Definition	0 pts—Absent/Not Applicable	1 pt—Partial	2 pts—Full/Mature	TRL/IRL/DT Basis
Y2	Response latency	No latency measurement, or the system does not operate during the incident; response time in minutes or unknown	System response time in the order of minutes (pre-computed with operator selection); acceptable for slowly developing fires	System response time $\leq$ tens of seconds (near-real time); consistent with the latency requirements of dynamic exit signage	TRL reactivity criterion: latency as a measure of operational readiness
Y3	Feedback loop	No feedback loop (open-loop): the system makes decisions but does not update the model based on action outcomes	Partial feedback: model updated once per incident or once per system state change; no continuous loop	Continuous closed-loop: action outcomes (e.g., ventilation change) are immediately fed back into the model and influence subsequent decisions	DT Maturity Levels 4–5 Zhuang et al., [26]; TRL adaptation criterion
Y4	Validation scope	Numerical simulations only (FDS, Pathfinder, and CFD) with no physical environment testing whatsoever	Laboratory testing or scaled physical model; validation in a controlled environment, but not in a full-scale building	Full-scale testing or deployment and operation in a real building; the results are available for external verification	TRL 4–5 (laboratory) vs. TRL 6–7 (operational environment) vs. TRL 8–9 (full deployment)
Y5	System autonomy	Operator required for every operational decision; the system functions exclusively as a human decision-support tool	The system proposes a route or action; the operator may accept or reject; the human approves every signage change	Full automation: the system independently updates signage and/or controls installations; the operator assumes a supervisory role only	IRL/SRL operational independence criterion

Source: The ordinal scale 0–2 is conceptually aligned with the TRL logic [23,24] for Dimension Y and the IRL logic [25] for Dimension X; references to the Digital Twin Maturity Model [26] are indicated in the "TRL/IRL/DT basis" column. Inter-rater reliability was assessed separately and is reported in Appendix B. Note: Background colours are used to distinguish the two assessment dimensions and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

Exact agreement was achieved in 48 of 60 assessments (80.0%); all 12 disagreements were by exactly one scale point. Following a joint discussion, the primary rater's original scores were retained in all 12 cases. Full inter-rater data (Table A2), the confusion matrix (Table A3), and kappa computation details (Table A4) are provided in Appendix B.

Rubric scores represent expert judgments based on explicit criteria (Table 3), not objective measurements.

- Replication by additional researchers applying the same Table 3 to the same six cases is possible and encouraged—discrepancies between raters would identify ambiguities in the scoring thresholds requiring refinement.
- The rubric requires further validation: (i) inter-rater validation with additional independent raters beyond the initial two-rater study reported in this manuscript; (ii) sensitivity analysis of the results to changes in threshold criteria; and (iii) application to a broader sample of DFS cases. This need for further refinement is consistent with broader guidance on exit modelling, which emphasises that the representation of

human behaviour remains a critical source of uncertainty in evacuation analysis and requires explicit modelling assumptions and careful interpretation [27].

- Rubric scores must not be treated as TRL certification or as an official assessment of technology readiness in the sense employed by the European Commission—they serve exclusively as a qualitative comparative tool within the scope of this review.

Table 4 presents the scoring results for the six cases based on the criteria defined in Table 3.

Table 4. The results of applying the DFS maturity assessment rubric to the six case studies (scoring criteria per Table 3).

Criteria (0–2 Scale)	Lille Wehbe	Beijing Ji	Taipei 101 Yen	Olympic Venues Liu	AIoT DT Tunnel Zhang	IDES Tunnel Wong
DIMENSION X—Technological Integration (Max. 10 pts)						
X1. Sensing (IoT/FDAS/CCTV)	0	2	2	2	2	1
X2. Building model (BIM/DT)	1	1	1	2	2	0
X3. Analytics/AI	2	1	2	1	2	1
X4. Decision algorithm	1	2	2	1	1	2
X5. Physical actuation	0	2	2	0	1	2
TOTAL X	**4**	**8**	**9**	**6**	**8**	**6**
DIMENSION Y—Operational Level (maximum 10 pt)						
Y1. System operational timing	1	2	2	2	2	2
Y2. Response latency	0	2	2	1	2	2
Y3. Feedback loop	0	1	1	1	2	0
Y4. Validation scope	1	1	0	1	1	1
Y5. System autonomy	0	2	2	1	1	2
TOTAL Y	**2**	**8**	**7**	**6**	**8**	**7**

Source: Authors' own elaboration—expert assessment by the primary rater (P.K.) based on technical descriptions in the cited publications. Detailed scoring criteria for the 0–2 scale are provided in Table 3. Inter-rater validation of the rubric was performed: $\kappa = 0.674$ [95% CI: 0.509, 0.839], $\kappa w = 0.797$. The primary rater's scores were retained after joint discussion. The bibliographic data from which individual scores are derived are discussed in Sections 4.1–4.6. Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

3.3. Methodological Limitations and Source Transparency

Limitations of the rubric: The DFS maturity assessment rubric (Tables 3 and 4) is an original instrument proposed in this paper. Its three principal limitations are: (1) inter-rater validation was subsequently performed by co-author W. Bonenberg, yielding unweighted Cohen's kappa = 0.674 [95% CI: 0.509, 0.839] and quadratic weighted kappa = 0.797, both indicating substantial agreement; the original scores were retained after joint discussion of all 12 disagreements; (2) subjectivity in the interpretation of technical descriptions in the cited publications—identical data may be interpreted differently by different raters given specific threshold criteria; and (3) limited sample size—six cases are insufficient to assess the statistical distribution of scores or to test the internal consistency of the scale. These limitations do not invalidate the rubric's utility as a comparative tool; however, they necessitate caution in extrapolating the findings.

Source transparency and data status differentiation. In this paper, three categories of claims are used, each carrying a distinct epistemological status that must be clearly distinguished:

- **Data verified by the authors in primary sources:** Liu et al. [28]—the results are reported in the cited document (route shortening of 17.2 m and 24.0 m); Ji et al. [29]—evacuation time data (−55 s compared to Dijkstra; −72% compared to static signage); Yen et al. [30]—evacuation success probability results (91%/61%/11%); and Wong et al. [31]—latency <0.1 s and the results of five tests. The authors had access to the full texts of these publications and directly verified the reported values.
- **Data cited from publications without independent computational verification:** Zhang et al. [9]—R^2 = 96.5% for fire location and HRR prediction. This value is reported in Section 4.2 of the cited paper Zhang et al. [9]; the present study did not replicate the calculations or experiments.
- **Expert evaluation by the authors: All scoring values in the proposed rubric, technology maturity values in the systemic gap analysis, and the positioning of cases within the conceptual comparison framework are based on expert evaluation conducted by the authors.** These values are derived from the interpretation and synthesis of information reported in the cited publications and are not taken directly from the original sources.

Absence of original calculations and simulations: This paper is a structured narrative review with case-based analysis—it does not include original fire experiments, FDS simulations, or evacuation drills. No quantitative meta-analysis was conducted due to the methodological heterogeneity of the cases analysed (different types of buildings, different definitions of the success of evacuation, and different test conditions). The contribution of this paper lies in qualitative synthesis and structured comparison—a recognised and appropriate approach within the structured narrative review with case-based analysis methodology, although one that requires explicit acknowledgement of the boundaries of inference. This approach is consistent with established fire safety risk index methods, in which hazard and protection-related factors are aggregated into a structured evaluative framework to support comparative decision-making [32]. An additional limitation concerns the heterogeneity of the case studies analysed, including differences in building types, evacuation conditions, and definitions of performance metrics. This limits the direct comparability of quantitative results and reinforces the need to interpret results as indicative rather than definitive. Furthermore, the absence of standardised evaluation protocols for DFSs introduces uncertainty in cross-case comparisons and highlights the need for future methodological harmonisation. Consequently, the findings should not be read as a prevalence estimate of DFS adoption in the global literature, but as a structured comparison of selected cases that illustrate the technological and regulatory conditions under which DFS-oriented solutions may emerge.

4. Results: Case Study Analysis

Table 5 presents the six DFS implementations together with the verification status of the reported evidence. Figure 3 synthesises the rubric-based assessment into a maturity gradient of DFS: panel (a) summarises the scores at the criterion-level, while panel (b) positions the six cases along the dimensions of technological integration and operational maturity. Figure 4 shows the coverage of the DFS layers in the analysed cases.

Figure 4 illustrates the coverage of the DFS layers in the cases analysed.

(a) DFS Maturity Assessment Rubric — 10 criteria, ordinal scale 0-2, max. 10 pts per dimension

Assessment criterion	Lille (Wehbe 2021)	Beijing (Ji 2022)	Taipei 101 (Yen 2024)	Olympic Venues (Liu 2020)	AIoT DT Tunnel (Zhang 2024)	IDES Tunnel (Wong 2025)
DIMENSION X — Technological Integration (X1-X5, max. 10 pts)						
X1. Sensing (IoT / FDAS / CCTV)	0	2	2	2	2	1
X2. Building model (BIM / DT)	1	1	1	2	2	0
X3. Analytics / AI	2	1	2	1	2	1
X4. Decision algorithm	1	2	2	1	1	2
X5. Physical actuation	0	2	2	0	1	2
TOTAL X (Technological Integration) — max. 10 pts	**4**	**8**	**9**	**6**	**8**	**6**
DIMENSION Y — Operational Level (Y1-Y5, max. 10 pts)						
Y1. System operational timing	1	2	2	2	2	2
Y2. Response latency	0	2	2	1	2	2
Y3. Feedback loop	0	1	1	1	2	0
Y4. Validation scope	1	1	0	1	1	1
Y5. System autonomy	0	2	2	1	1	2
TOTAL Y (Operational Level) — max. 10 pts	**2**	**8**	**7**	**6**	**8**	**7**

0 — absent / not applicable 1 — partial / limited 2 — full / mature

(b) DFS Maturity Gradient — six case studies on Dimension X (IRL logic) and Dimension Y (TRL logic) axes, based on rubric scores from panel (a)

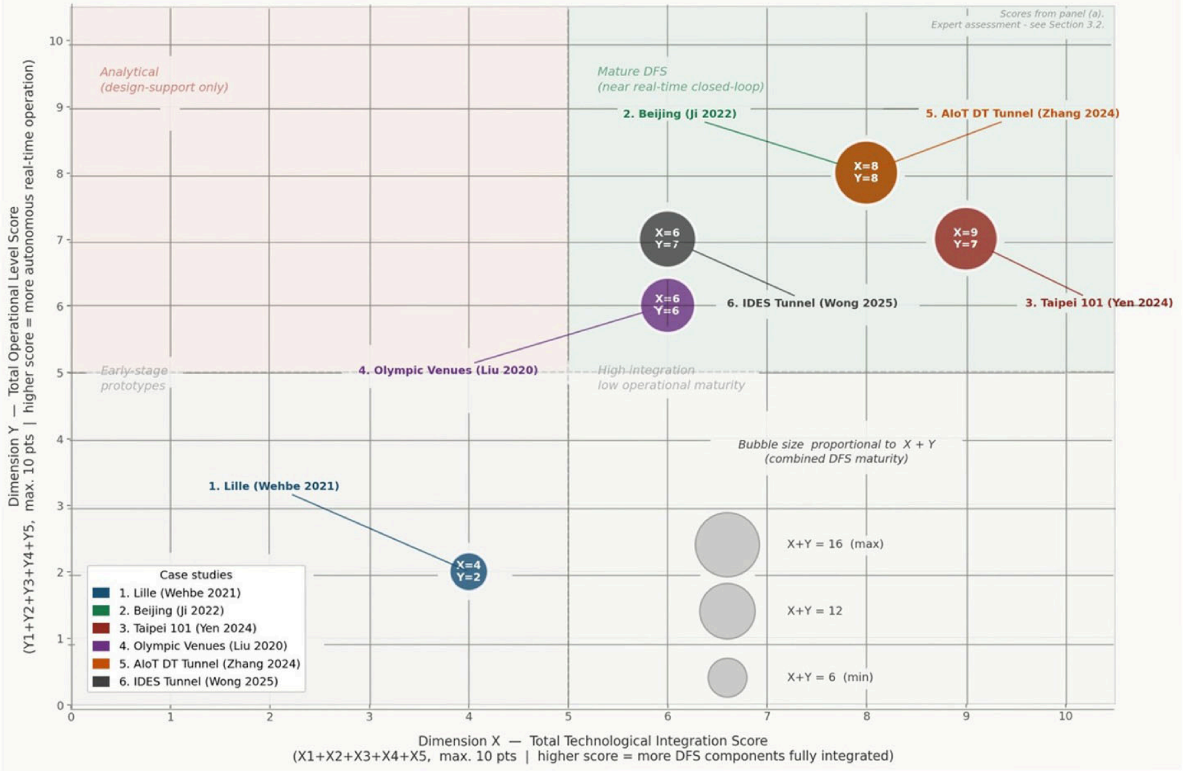


Figure 3. DFS maturity gradient in six case studies: (**a**) criterion-level rubric scores based on Table 4 and interpreted according to the thresholds defined in Table 3; (**b**) position of the cases along Dimension X (technological integration/IRL logic) and Dimension Y (operational maturity/TRL logic). Inter-rater validation: κ = 0.674, κw = 0.797. Authors' own elaboration.

Table 5. Comparison of six DFS implementations [the "Benefits" column includes verification status indicators].

No.	Case Study/Source	Building Type (ZL Analogue)	Key DFS Components	Maturity Level	Reported Benefits [Verification Status]	Main Limitations
1	Wehbe et al.—Lille	University building/laboratory (ZL I analogue)	BIM + FDS + Pathfinder + pre-computed scenario database + AI/ML platform	Analytical—decision support	Improved exit utilisation; avoidance of tenability-compromised zones; RSET < ASET criterion satisfied [data from primary source, cited]	No real-time sensors; no dynamic exit signage; pre-computed scenarios only
2	Ji et al.—Beijing	Medical facility (ZL II analogue)	BIM + IoT LoRa/STM32 + CCTV/AI + DCA algorithm + dynamic LED exit signs	Near-real-time—executive layer	Evacuation time reduced by 55 s vs. the Dijkstra algorithm; up to 72% reduction vs. static signage [data verified by authors in primary source]	Simplified fire model; single-floor analysis only
3	Yen and Lin—Taipei 101	Shopping mall (ZL III analogue)	IoT + FDS 7.9.1 + ESP algorithm + IIFESS system + dynamic exit signage	Scalable real-time, multi-floor	+34 percentage points in the evacuation success rate; simulation data only [data verified by authors in primary source]	No physical deployment; simplified congestion model; no experimental validation
4	Liu et al.—Winter Olympic Venues	Large sports hall (ZL I analogue)	BIM + IoT + Digital Twin + modified Dijkstra algorithm	Real-time—DT prototype	Evacuation route reduced by 17.2 to 24.0 m in hazard scenarios [data verified by the authors in the primary source]	No dynamic LED signage; no full closed-loop; simulation-based validation
5	Zhang et al.—AIoT DT Tunnel	Road tunnel (critical infrastructure)	AIoT—dense sensor network + living Digital Twin + Transformer model (AI)	Near-real-time closed-loop—framework/lab stage	R2 = 96.5% for fire location and HRR prediction (reported in a primary source, cited); three full-scale fire tests	No deployment in the operational tunnel; wireless network reliability issues
6	Wong et al.—IDES Tunnel	Tunnel model, 1:50 scale (demonstration)	Thermocouple network (ESP32 + MQTT) + dynamic LED exit signs + reversion mechanism	Real-time—actuation layer	Latency <0.1 s; 100% correct sign patterns; reversion mechanism operational [data verified by authors in the primary source]	Laboratory scale of 1:50; no occupant behaviour study; no full-scale validation

Source: Authors' own elaboration based on the six case studies discussed in Sections 4.1–4.6. Notes: [data verified by the authors in the primary source] = authors had access to the full text and checked the reported values; [reported from primary source, cited] = values reported as per the cited publication, without independent computational verification; and [simulation data only] = numerical results only, without physical experimentation. Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

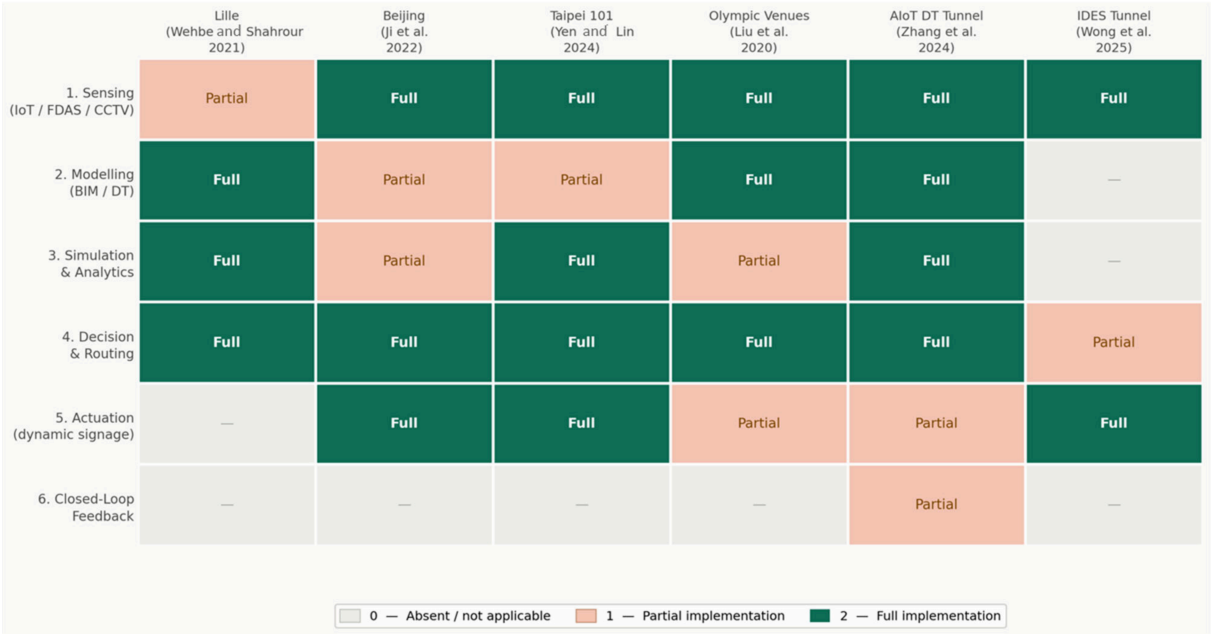


Figure 4. Coverage of DFS layers across the six case studies. Authors' own elaboration.

4.1. Lille—Wehbe et al. [33]

Laboratory LGCgE, Université de Lille (~1256 m^2, ~50 occupants): BIM + FDS + Pathfinder + pre-computed scenario database. Data from the primary source (cited): ASET ≈ 27 s for Exit 1; evacuation time 54.5 s (without fire) vs. 251.5 s (with fire). Rubric scores (per Table 4): X = 4 (X1 = 0—no real-time sensors; X5 = 0—no physical action; X2 = 1—static BIM); Y = 2 (Y1 = 1—operator decision support; Y2 = 0—no latency measurement; Y3 = 0—open-loop; Y4 = 1—simulations with limited validation; and Y5 = 0—operator required for all decisions). Approximate TRL equivalent: 4–5.

4.2. Beijing Capital Airport Emergency Center—Ji et al. [29]

Airport emergency center (29 rooms, corridor layout): BIM + IoT LoRa/STM32 + CCTV/AI + DCA algorithm + dynamic LED exit signs. **Data verified by the authors in the primary source:** evacuation time reduced by 55 s vs. the Dijkstra algorithm; up to 72% reduction compared with static signage (with 14 evacuees, and a fire source blocking a corridor). Rubric scores (per Table 4): X = 8 (X1 = 2, X2 = 1, X3 = 1, X4 = 2, X5 = 2); Y = 8 (Y1 = 2, Y2 = 2, Y3 = 1, Y4 = 1, Y5 = 2). Approximate TRL equivalent: 7.

4.3. Taipei 101 Shopping Mall—Yen et al. [30]

Three-story shopping mall: IoT + FDS 7.9.1 + ESP algorithm + IIFESS system. **Data verified by the authors in the primary source (simulation data only—no physical deployment)**, with 2000 evacuees, a methane fire scenario, no sprinklers: ESP = 91%, FEL = 61%, RS = 11% (DFS vs. FEL: +34 percentage points). Rubric scores (per Table 4): X = 9 (X1 = 2, X2 = 1, X3 = 2, X4 = 2, X5 = 2); Y = 7 (Y4 = 0—absence of physical validation reduces the score). The approximate TRL equivalent was 5–6, reflecting high technological integration without physical deployment.

4.4. Winter Olympic Venues—Liu et al. [28]

Yanqing bobsleigh and luge venue support building: BIM + IoT LoRa + UWB localisation + modified Dijkstra algorithm + Digital Twin visualised via WebGL. Data were verified by the authors in the primary source (Table 3 of the article cited): the evacuation route was shortened by 17.2 m (8.6 s) for room 2R5 and by 24.0 m (12.0 s) for room 2R4, compared

to the unoptimised route. No dynamic LED signage (actuation via mobile application). Rubric scores: X = 6 (X5 = 0); Y = 6. Approximate TRL equivalent: 6.

4.5. AIoT Digital Twin Tunnel—Zhang et al. [9]

SCFRI laboratory tunnel, Chengdu (140 × 8 × 6 m): ESP8266 + MQTT + Transformer model + Unity 3D/WebGL platform. **Data were cited from the primary source without independent computational verification:** R^2 = 96.5% for fire location and HRR prediction (as reported in Zhang et al. [9]). Three full-scale fire tests (pool fires + wood crib) were conducted by the authors of the cited work; there was no deployment in an operational road tunnel. Rubric scores: X = 8; Y = 8 (Y4 = 1—full-scale tests conducted under laboratory tunnel conditions). Approximate TRL equivalent: 6–7.

4.6. IDES Tunnel—Wong et al. [31]

Physical tunnel model on a 1:50 scale (1700 × 170 × 140 mm): Four thermocouples (ESP32 + MQTT) + three dynamic LED exit signs + reversing mechanism. **Data were verified by the authors in the primary source (Table 2 of the article cited)**: sign pattern change latency <0.1 s from threshold (58 °C) exceedance; 100% correct sign patterns across five tests; and the eversion mechanism activated in Test 2D-b (t = 25 s). Occupant behaviour was not studied. Rubric: X = 6 (X2 = 0; X3 = 1); Y = 7. The approximate TRL equivalent was 6–7 for the actuation layer. The absence of a building model layer (X2 = 0) has direct implications for system reliability: the IDES system operates as an open-loop actuation chain in the modelling dimension, with sign decisions driven exclusively by threshold exceedance in a fixed sensor-to-sign mapping, without any spatial model of tunnel geometry, fire propagation, or occupant distribution. This means the same sign pattern is activated regardless of fire location, scale, or propagation direction, constraining routing adaptability. Scaling to an operational tunnel would require, at a minimum, a static geometric model (X2 $\geq$ 1) to enable location-aware routing decisions. The IDES case thus confirms the IRL logic of the rubric: high actuation capability (X5 = 2) without modelling integration does not constitute a mature DFS solution.

The IDES Tunnel case Wong et al. [31]) achieved a relatively high actuation score but a lower technological integration score (X = 6; X2 = 0, X3 = 1), mainly due to the complete absence of a building model and limited analytics. This highlights an important architectural insight: while a geometric digital model significantly improves the reliability and decision accuracy—enabling location-aware routing and feedback-driven adaptation—a lightweight and actuation-focused system can still deliver strong operational performance (Y = 7) in terms of sign-change latency (<0.1 s) and automatic reversion logic. The IDES case thus demonstrates that different DFS maturity profiles may be appropriate for different operational contexts and resource constraints: a full DFS stack (X $\geq$ 8, Y $\geq$ 7) is warranted for complex multi-floor buildings, whereas a simplified actuation layer may be a viable first implementation step for linear infrastructure, such as road tunnels.

4.7. Cross-Case Synthesis

The rubric (Tables 3 and 4; Figure 3) confirms a maturity gradient of X = 4–9, Y = 2–8. Three analytically significant observations emerge:

Physical action (X5) appears to be the strongest predictor of Dimension Y within the analysed cases: its absence directly reduces Y1 and Y5 (IRL logic—without C2P loop, the system cannot be operationally functional).

Closed-loop feedback (Y3) is the least implemented layer—scoring 0 or 1 in five of the six cases—corresponding to Levels 3–4 of Zhuang's DT Maturity Model.

X–Y discrepancy (Taipei 101, X = 9 with Y = 7): high technological integration does not guarantee operational maturity in the absence of physical validation (Y4 = 0). This is a methodologically significant finding: it highlights the risk of overestimating systems validated exclusively through simulation.

These cross-case findings also clarify the sustainability relevance of DFS-oriented systems across the six analysed cases. Their value lies not only in improving evacuation outcomes during an emergency, but also in strengthening the long-term resilience of building operation. By allowing fire safety logic to be updated through sensor data, digital models, scenario-based evaluation, and, in more mature cases, physical actuation, DFS can reduce the mismatch between design stage assumptions and actual operational conditions. This supports a more adaptive and lifecycle-oriented understanding of building safety, which is particularly important in facilities where functions, occupancy patterns, and risk profiles change over time.

5. Discussion

5.1. Responses to Research Questions

RQ1: The sensor decision chain $\rightarrow$ is present in all six cases; it is fully operationally closed without human intervention only in the Beijing case [29] and demonstrably in Zhang et al. [9].

RQ2: The minimum functional DFS pattern requires: Sensing (X1 $\geq$ 1) + Analysis (X3 $\geq$ 1) + Decision (X4 $\geq$ 1) + Actuation (X5 $\geq$ 1). BIM is present in all six cases as the geometric foundation; a living DT (X2 = 2) is present in three.

RQ3: The six cases suggest that the transposition of selected DFS principles into the Polish regulatory context may be technically plausible for selected complex facilities corresponding typologically to ZL I–III occupancy categories. The barrier is regulatory: there is no equivalence pathway for DFS in Polish law (Gap P1). The rubric (Table 4) further demonstrates that none of the six cases achieved Y4 = 2 (complete scale validation in an operational building), which would represent the minimum threshold for formal regulatory acceptance. However, this conclusion should be understood as analytical transferability rather than statistical generalisability. The six cases provide typological analogues for selected ZL I–III occupancy categories and complex infrastructure, but they do not represent the full diversity of all ZL I–III buildings in Poland. Therefore, the proposed regulatory recommendations should be treated as a cautious framework for further pilot implementation and validation, rather than as evidence that DFS solutions are universally applicable across all ZL I–III facilities. The main barrier remains regulatory: there is currently no explicit equivalence pathway for DFS in Polish law.

5.2. DFS Effectiveness: Data and Their Evidential Status

Figure 5 presents the quantitative results with the verification status of each data point explicitly indicated. Data from the Beijing case [29] and the Winter Olympic venues case [28] were verified by the authors in primary sources. The Taipei 101 results are simulation-only and therefore must be interpreted with greater caution. Nevertheless, the advantage of dynamic evacuation guidance over static signage is consistent with the broader findings reported in the literature [10]. The result reported by Zhang et al. [9] is cited without independent computational verification.

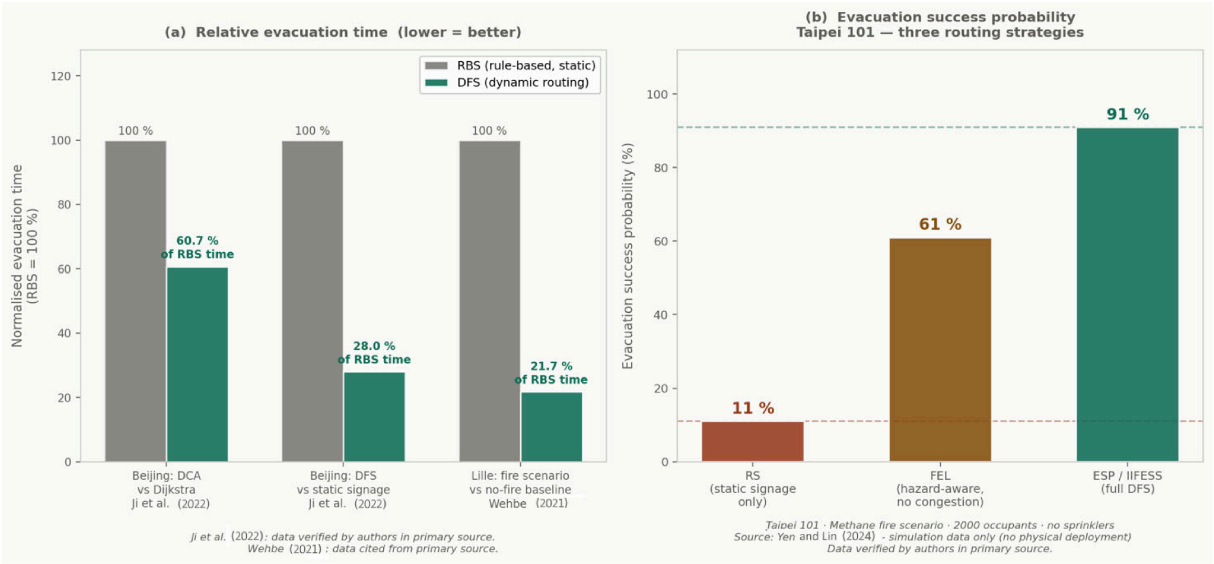


Figure 5. Comparison of evacuation performance under RBS and DFS strategies: (**a**) relative evacuation time, with RBS = 100%; (**b**) evacuation success probability in the Taipei 101 case, comparing three routing strategies under a methane fire scenario with 2000 occupants. Beijing data were verified by the authors in the primary source; Taipei 101 data are simulation-only and involve no physical deployment. Authors' own elaboration.

5.3. Systemic Gaps and Regulatory Map

Table 6 and Figure 6 map six systemic gaps. The technology maturity values in column "0–10" represent the authors' expert assessment—not the result of a formal TRL audit. The gap P1b (dynamic exit signage, maturity = 7/10) concerns the component with the highest technological maturity (X5 = 2, TRL equivalent ~7–8 per Table 6)—technology is ready; however, regulation has not yet responded. This broader mismatch between technical possibilities and fire safety practice is consistent with the recent review literature that highlights unresolved structural fire hazards, regulatory challenges, and the need for more advanced prevention strategies [34]. A related gap concerns the continuity of BIM/DT throughout the building lifecycle: this problem is consistent with the broader literature, which shows that the implementation of BIM in existing buildings remains constrained by the high effort of model creation, the difficulty of updating the information, and the uncertainty of available building data [2]. Both axes represent expert assessments by the authors and do not correspond to formal TRL or FMEA evaluations.

Table 6. Systemic gap matrix: RBS vs. PBD/DFS.

Systemic Gap	Category	Current Status (RBS)	DFS Requirement	Tech. Maturity (0–10)	Priority
Absence of a formal PBD pathway in Polish regulations	Legal (PL)	The technical conditions and MoI regulations define only prescriptive requirements	Equivalence procedure for DFS vs. RBS; approval authority; acceptance criteria	3/10	P1—Critical
Legal status of dynamic exit signage	Tech-reg	EN 1838/PN-EN 50172 mandate static signage	Definition of "active evacuation sign"; fail-safe standards; testing protocols	7/10	P1—Critical

Table 6. *Cont.*

Systemic Gap	Category	Current Status (RBS)	DFS Requirement	Tech. Maturity (0–10)	Priority
BIM/DT outside the building lifecycle	Tech-reg	BIM mandatory only for public procurement above the threshold; no operational update obligation	BIM as a living fire safety record; DT as a DFS prerequisite	6/10	P2—High
Absence of AI/DCA algorithm validation standards	Evidential	No certification standards for decision algorithms in fire safety systems	Scenario test protocol (FDS + evacuation); ML acceptance criteria	4/10	P2—High
Independent verification of simulations	Evidential	FDS/Pathfinder accepted without external validation	Benchmark case database; independent FSE expert reviewer	5/10	P3—Moderate
Human factors and signage ergonomics	Evidential	Regulations specify only physical sign parameters; no cognitive effectiveness studies	Behavioural studies on response times; legibility standards under stress	4/10	P4—Monitored

Source: Authors' own elaboration. Technology maturity and impact values represent expert assessments based on qualitative triangulation of three inputs: rubric scores (Table 4), performance data reported in the analysed primary studies, and regulatory gap analysis. These values are not the result of a formal TRL audit or quantitative FMEA assessment. Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.



Figure 6. Systemic gap map: technological maturity versus impact on evacuation safety. Both axes represent authors' expert assessments based on qualitative triangulation of rubric scores, reported performance evidence, and regulatory gap analysis; they do not correspond to formal TRL or FMEA evaluations. Authors' own elaboration.

The map distinguishes four decision zones, highlighting priorities for regulatory intervention versus research needs.

5.4. EU and Polish Regulatory Framework

European regulations—EU Directive 2004/54/EC (road tunnels), Construction Products Regulation CPR 2024/3110, and EN 50172—remain predominantly prescriptive. No formal acceptance pathway exists for DFS components (living Digital Twin, AI-driven dynamic exit signage, and closed-loop systems) as a method of demonstrating equivalence with regulatory requirements. This problem is consistent with recent research in Germany, which shows that prescriptive building regulations require interpretation through safety objectives and residual-risk reasoning whenever building types or deviations are not fully covered by detailed code provisions [35]. This regulatory gap is particularly problematic in complex and continuously operating facilities, where performance-based fire strategies can offer safer and less disruptive alternatives to strictly prescriptive solutions [17]. This structural discrepancy—most acute in critical infrastructure, such as road tunnels—means that technically superior evidence-supported solutions remain outside the mainstream of design approval practice.

5.5. Human Factors and Research Gaps

Wong et al. [31] (data verified by the authors) investigated exclusively technical reliability (Y4 = 1 in the rubric). Empirical data are not available on occupant behaviour in response to dynamic exit signage under actual fire conditions. This constitutes Gap P4 in Table 6 and should be prioritised in future research, particularly in buildings with mobility-impaired occupants (ZL II occupancy).

6. Conclusions and Recommendations

6.1. Conclusions

DFS can improve evacuation resilience under specific technological and operational conditions identified in the analysed cases. The formal rubric (Table 4) confirms a maturity gradient of X = 4 to 9, Y = 2 to 8. Data verified directly by the authors in primary sources confirm DFS benefits in three of the six cases.

The maturity gradient suggests an incremental implementation pathway: the physical actuator layer (X5) first, followed by the living DT (X2), and finally closed-loop feedback (Y3).

Dynamic exit signage (X5) delivers the greatest operational benefits at the highest maturity of TRL (~7–8), yet it lacks legal status in the regulatory frameworks of Poland and the EU (Gap P1b).

Closed-loop feedback (Y3) is the critically underdeveloped layer—corresponding to Levels 3 to 4 of Zhuang's DT maturity model in all six cases—and is the key to achieving TRLs 8 to 9 for DFS.

The rubric requires further validation. The scores in Table 4 were independently verified by co-author W. Bonenberg. The agreement between the authors was substantial: Cohen's kappa = 0.674 [95% CI: 0.509, 0.839] and quadratic weighted kappa = 0.797. The primary rater's scores were retained after joint discussion. Further validation with additional raters is recommended before the rubric is adopted as a formal benchmarking standard.

The findings also strengthen the sustainability dimension of dynamic fire safety systems. In this study, sustainability is understood not only as environmental efficiency, but also as the capacity of buildings and infrastructure to remain safe, resilient, and operationally adaptable throughout their life cycle. DFS contributes to this broader understanding by integrating real-time hazard information, digital modelling, adaptive decision-making, and, where technologically mature, dynamic actuation into a single safety-

management logic. Consequently, BIM/DT/IoT/AI-based fire safety systems should be interpreted as components of sustainable building performance, particularly in complex facilities where user safety, operational continuity, and adaptability to uncertain future scenarios are critical.

6.2. Normative Recommendations

[P1-REC1] Formal PBD pathway: Amendment of the Polish Technical Conditions and relevant ministerial regulations to introduce an explicit DFS equivalence mechanism, modelled on BS 9999 (UK) or NFPA 101 (USA).

[P1-REC2] Legal status for dynamic exit signage: Amendment of PN-EN 1838 and EN 50172 to introduce the category of "active evacuation sign" with reliability (fail-safe) and testing requirements; a CEN standardisation initiative at the EU level.

[P2-REC3] BIM/DT within the building lifecycle: A mandatory BIM update obligation when the function of ZL I–III buildings is altered; longer-term: DT as a component of technical construction requirements.

[P2-REC4] AI algorithm validation protocol: Development of a scenario test protocol (FDS + evacuation simulation + dynamic signage tests) with acceptance criteria for ML algorithms in safety-critical fire-protection systems. The practical feasibility of this recommendation must acknowledge that current EU and Polish legal frameworks do not yet provide a dedicated certification pathway for non-deterministic or probabilistic algorithms in life safety applications. The EU AI Act provides a broader regulatory context for high-risk AI systems, but it does not by itself establish a fire safety-specific approval procedure for DFS routing algorithms. Relevant analogues exist in other regulated safety-critical sectors, including functional automotive safety and medical device software; in the Polish context, CNBOP-PIB is identified as the most appropriate body to conduct type-approval certification for DFS algorithmic components. Until such a framework exists, any DFS deployment relying on non-deterministic ML-based routing should be approved through an individual equivalence assessment under the performance-based pathway proposed in P1-REC1. It should be noted, however, that P2-REC4 is a forward-looking normative recommendation rather than a direct empirical finding derived from all six case studies. The analysed cases demonstrate the increasing role of algorithmic decision-making in DFS, but they do not provide a complete validation protocol for safety-critical AI. Therefore, the proposed AI validation pathway should be interpreted as a regulatory requirement inferred from the identified gap between technological capability and approval mechanisms, rather than as a procedure already validated within the case-study sample.

[REC5] Pilot projects: Pilot DFS implementations in selected representative ZL occupancy categories (university, hospital, and shopping centre) under Polish conditions involve the National Headquarters of the State Fire Service (KG PSP) and the Scientific and Research Centre for Fire Protection–National Research Institute (CNBOP-PIB) as validation institutions. These findings should be interpreted as a contribution to the ongoing development of performance-based fire safety approaches rather than as definitive evidence of universal applicability.

6.3. Directions for Future Research

Recent studies on emergency Digital Twins and IoT-based fire emergency management further confirm the importance of multimodal sensor fusion for improving fire localisation accuracy, real-time situational awareness, and evacuation decision latency [36,37].

Recent research indicates that future DFS development should move toward more integrated, data-rich, and cyber–physical environments. AIoT-powered building Digital Twins are increasingly being proposed for smart firefighting and super-real-time fire

forecasting [38], while metaverse-based evacuation training systems supported by deep reinforcement learning may improve preparedness and occupant response under emergency conditions [39]. BIM- and computer-vision-based evacuation frameworks also demonstrate the importance of localisation systems and real-time spatial awareness for fire emergency evacuation [40]. In parallel, informative Digital Twins and Digital Twin-enabled evacuation models are being developed to support fire-risk assessment, individual evacuation decisions, and broader building resilience [41,42]. Future DFS research should therefore address semantic interoperability between BIM/IFC models and mobile crowd-sensing data [43], automated fire safety compliance checking in operational buildings [44], and the regulatory transfer of responsive fire safety approaches across different legal regimes [45]. GIS- and IoT-supported three-dimensional evacuation frameworks may further strengthen real-time route planning in complex facilities [46]. Finally, the development of DFS should remain connected to broader fire safety regulation and hazard-mitigation strategies, including lessons from national regulatory reviews and general fire hazard assessment frameworks [47,48].

Inter-rater rubric validation: Independent application of Tables 3 and 4 by a second researcher to the same six cases and inter-rater agreement analysis (Cohen's kappa coefficient).

- **Human factors (Gap P4):** Behavioural studies on the response of occupants to dynamic exit signage under stress conditions, specifically targeting mobility-impaired building users.
- **Cybersecurity:** Vulnerability analysis of IoT + AI systems in fire safety-critical applications.
- **Regulatory transfer:** Comparative analysis of PBD/DFS approval approaches in Germany, the Netherlands, and France to identify best practice frameworks for adaptation in Polish legislation.

Author Contributions: Conceptualisation, P.K. and W.B.; methodology, P.K.; formal analysis, P.K.; investigation, P.K.; resources, P.K. and B.K.; writing—original draft preparation, P.K.; writing—review and editing, W.B., B.K. and A.S.-G.; visualisation, P.K. and A.S.-G.; supervision, W.B., A.S.-G., B.K. and R.P. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

AI—Artificial Intelligence; AIoT—Artificial Intelligence of Things; ASET—Available Safe Egress Time; BIM—Building Information Modelling; BMS—Building Management System; CAD—Computer-Aided Design; CFD—Computational Fluid Dynamics; DFS—Dynamic Fire Safety System; DT—Digital Twin; ESP—Evacuation Safety Percentage; FAS—Fire Alarm System; FDS—Fire Dynamics Simulator; FEL—Part of Evacuees Lost; FMEA—Failure Mode and Effects Analysis; FSE—Fire Safety Engineering; IoT—Internet of Things; IRL—Integration Readiness Level; ML—Machine Learning; PBD—Performance-Based Design; RBS—Rule-Based Safety; RSET—Required Safe Egress Time; SRL—System Readiness level; TRL—Technology Readiness Level; VAS—Voice Alarm System; and UWB—Ultra-Wideband.

Appendix A. Literature Search Documentation

This appendix documents the six Scopus query strings used in the structured narrative review with case-based analysis (Section 3.1) and the case study selection flow from initial retrieval to the six included case studies. It is provided as a transparency measure in response to reviewer feedback on search reproducibility.

Table A1. Scopus query strings used in the structured narrative review (database: Scopus; search date: May 2026; language filter: English; document types: article, review, conference paper).

No.	Search Purpose	Scopus Query String (TITLE-ABS-KEY)	Records Retrieved	Records Included (IN)
S1	DFS core—BIM/DT + evacuation (narrowest, highest precision)	TITLE-ABS-KEY ((“dynamic fire safety” OR “dynamic fire safety system” OR “DFS”) AND (“BIM” OR “building information model *” OR “digital twin*”) AND (“evacuation” OR “egress”)) AND DOCTYPE(ar OR re OR cp) AND PUBYEAR > 2018	**20**	**8**
S2	DFS components—IoT/AI/real-time	TITLE-ABS-KEY((“fire safety” OR “fire evacuation” OR “emergency evacuation”) AND (“BIM” OR “digital twin*”) AND (“IoT” OR “internet of things” OR “artificial intelligence” OR “machine learning”) AND (“case stud*” OR “simulation” OR “real-time”)) AND DOCTYPE(ar OR re OR cp) AND PUBYEAR > 2018	**20**	**6**
S3	Responsive/smart architecture + fire safety + sensors	TITLE-ABS-KEY((“responsive architecture” OR “adaptive building*” OR “smart building*”) AND (“fire safety” OR “fire protection” OR “evacuation”) AND (“BIM” OR “digital twin*” OR “IoT” OR “sensor*”)) AND DOCTYPE(ar OR re OR cp) AND PUBYEAR > 2015	**46**	**11**
S4	Paradigm shift RBS → PBD—regulatory and theoretical foundation	TITLE-ABS-KEY((“performance-based design” OR “performance-based fire” OR “PBD”) AND (“prescriptive” OR “rule-based”) AND (“fire safety” OR “fire engineering”)) AND DOCTYPE (ar OR re) AND PUBYEAR > 2015	**43**	**10**
S5	Dynamic exit signage—physical actuation layer (Note: 72% noise from transport domain)	TITLE-ABS-KEY((“dynamic evacuation sign*” OR “intelligent evacuation sign*” OR “adaptive sign*” OR “dynamic guidance”) AND (“fire” OR “emergency” OR “evacuation”) AND (“IoT” OR “sensor*” OR “real-time” OR “algorithm*”)) AND DOCTYPE(ar OR re OR cp) AND PUBYEAR > 2018	**18**	**3**
S6	Digital Twin + fire safety/emergency in buildings (broadest; partially overlaps S1–S2)	TITLE-ABS-KEY((“digital twin*”) AND (“fire safety” OR “fire detection” OR “evacuation” OR “emergency management”) AND (“building*” OR “facility” OR “infrastructure”)) AND DOCTYPE(ar OR re OR cp) AND PUBYEAR > 2019	**104**	**14** [1]
TOTAL (unique)		After deduplication across all 6 strings; S5 transport noise removed; S6 overlap with S1–S2 accounted for	**≈147 raw ≤81 unique**	**≤27** [2]

Notes in Table A1. [1] S6 retrieved 104 records, of which 30 (29%) are out-of-scope (wildfire, flood, agriculture, seismic, and traffic management) and ~12 are duplicates of S1–S2. Net new IN contribution from S6: ~14 records. [2] ≤27 records passed title/abstract screening; ~9 were subsequently excluded at full-text stage, yielding 6 included case studies. S5 retrieved 18 records, of which 13 were traffic management false hits (72% noise); three included records were confirmed in primary sources. Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.



Figure A1. Case study selection flow—from Scopus retrieval to six included case studies (structured narrative review, PICo protocol; Torgerson [20]). Figure A1 note. The selection reflects purposive rather than exhaustive sampling Torgerson, 2003 [20]. The objective was to represent the full spectrum of DFS maturity (X = 4–9; Y = 2–8) across building types analogous to Polish occupancy categories ZL I–III and critical infrastructure. All six included cases were verified in primary sources by the authors (Section 3.3). Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated [9,28–31,33].

Appendix B. Inter-Rater Reliability Data for the DFS Maturity Assessment Rubric

Rater 1 (P.K.) = Przemysław Konopski (primary rater, manuscript author). Rater 2 (W.B.) = Wojciech Bonenberg (co-author; independent scoring, blind to primary scorer scores). Assessment date: May 2026. Scoring criteria as defined in Table 3 of the main text.

Table A2. Independent scores of P.K. and WBC in all 60 case assessments based on criteria (10 criteria × 6 case studies, ordinal scale 0–2). Orange cells = disagreement (Δ = 1; n = 12). TOTAL rows: amber value = differing dimension sum. Scoring criteria according to Table 3.

Criteria (Scale 0–2)	C1 Lille Wehbe [33]	C2 Beijing Ji [29]	C3 Taipei 101 Yen [30]	C4 Olympic Liu [28]	C5 AIoT Zhang [9]	C6 IDES Wong [31]
DIMENSION X—Technological Integration (X1–X5)						
X1. Sensing (IoT/FDAS/CCTV)	**PK = 0** WB = 0	**PK = 2** **WB = 1 Δ = 1**	**PK = 2** WB = 2	**PK = 2** WB = 2	**PK = 2** **WB = 1 Δ = 1**	**PK = 1** WB = 1
X2. Building model (BIM/DT)	**PK = 1** WB = 1	**PK = 1** **WB = 2 Δ = 1**	**PK = 1** WB = 1	**PK = 2** WB = 2	**PK = 2** WB = 2	**PK = 0** WB = 0
X3. Analytics/AI	**PK = 2** WB = 2	**PK = 1** WB = 1	**PK = 2** **WB = 1 Δ = 1**	**PK = 1** WB = 1	**PK = 2** WB = 2	**PK = 1** WB = 1
X4. Decision algorithm	**PK = 1** WB = 1	**PK = 2** WB = 2	**PK = 2** WB = 2	**PK = 1** WB = 1	**PK = 1** WB = 1	**PK = 2** WB = 2
X5. Physical actuation	**PK = 0** WB = 0	**PK** = 2 WB = 2	**PK = 2** **WB = 1 Δ = 1**	**PK = 0** **WB = 1 Δ = 1**	**PK = 1** WB = 1	**PK = 2** WB = 2
TOTAL X (max. 10 pts)	**PK = 4** WB = 4	**PK = 8** WB = 8	**PK = 9** WB = 7	**PK = 6** WB = 7	**PK = 8** WB = 7	**PK = 6** WB = 6
DIMENSION Y—Operational Level (Y1–Y5)						
Y1. System operational timing	**PK = 1** **WB = 0 Δ = 1**	**PK = 2** WB = 2	**PK = 2** WB = 2	**PK = 2** **WB = 1 Δ = 1**	**PK = 2** WB = 2	**PK = 2** WB = 2
Y2. Response latency	**PK = 0** WB = 0	**PK = 2** WB = 2	**PK = 2** WB = 2	**PK = 1** WB = 1	**PK = 2** WB = 2	**PK = 2** WB = 2
Y3. Feedback loop	**PK = 0** WB = 0	**PK = 1** **WB = 2 Δ = 1**	**PK = 1** WB = 1	**PK = 1** WB = 1	**PK = 2** WB = 2	**PK = 0** **WB = 1 Δ = 1**
Y4. Validation scope	**PK = 1** WB = 1	**PK = 1** WB = 1	**PK = 0** WB = 0	**PK = 1** WB = 1	**PK = 1** WB = 1	**PK = 1** WB = 1
Y5. System autonomy	**PK = 0** WB = 0	**PK = 2** WB = 2	**PK = 2** WB = 2	**PK = 1** **WB = 2 Δ = 1**	**PK = 1** WB = 1	**PK = 2** **WB = 1 Δ = 1**
TOTAL Y (max. 10 pts)	**PK = 2** WB = 1	**PK = 8** WB = 9	**PK = 7** WB = 7	**PK = 6** WB = 6	**PK = 8** WB = 8	**PK = 7** WB = 7

Note. Exact agreement: 48/60 assessments (80.0%). All 12 disagreements are off by 1; no off-by-2 disagreement occurred. Following joint discussion, P.K.'s scores were retained in all 12 cases (see Table A5 for resolution notes). Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

Table A3. 3 × 3 confusion matrix for Cohen's calculation κ (PK rows × W.B. columns; n = 60). Green diagonal cells = exact agreement (7 + 19 + 22 = 48); white off-diagonal cells = one-point disagreements (12 total; no two-point disagreements).

P.K. Score\W.B. Score	WB = 0	WB = 1	WB = 2	Row Total
PK = 0	**7**	2	0	**9**
PK = 1	1	**19**	3	**23**
PK = 2	0	6	**22**	**28**
Total columns	**8**	**27**	**25**	**60**

Kappa computation: From Table A3, observed agreement p_o = 48/60 = 0.800; chance-expected agreement p_e = 0.387 (marginal products). Unweighted κ = (0.800 − 0.387)/(1 − 0.387) = 0.674. Standard error SE = 0.0932; 95% CI = [0.509, 0.839]. Quadratic weighted κw = 0.797 (penalises off-by-2 disagreements 4× more than off-by-1 disagreements; recommended primary metric for ordinal 0–1–2 scales [49]). Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

Table A4. Summary of kappa values with interpretation according to Landis and Koch [50].

Kappa Variant	Weighting Logic	Value	95% CI	Interpretation (Landis & Koch 1977) [50]
Unweighted Cohen's κ	All disagreements are penalised equally	**κ = 0.674**	[0.509, 0.839]	**Substantial**
Linear weighted κw	Off-by-1 penalised half of off-by-2	**κw = 0.729**	—	**Substantial**
Quadratic weighted κw (recommended for ordinal scales)	Off-by-1 penalised 4× less than off-by-2	**κw = 0.797**	—	**Substantial (→ almost perfect)**

Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

Table A5. Resolution notes for all 12 discrepancies in the assessments. Green = P.K. score (retained); amber = W.B. score. All 12 cases were resolved by retaining the P.K. score after joint discussion.

Crit.	Case	P.K.	W.B.	Resolution	Interpretation Note and Basis for Retaining the P.K. Score
X1	C2 Beijing	**2**	**1**	**P.K. retained**	W.B.: Assessed LoRa/STM32 as partial sensing. P.K.: Counted IoT + CCTV + AI combination as full (continuous geolocated streams). Criteria met.
X1	C5 AIoT Tunnel	**2**	**1**	**P.K. retained**	W.B.: Lab-scale sensor density = partial. P.K.: Continuous geolocated ESP8266 streams satisfy the criterion regardless of deployment scale.
X2	C2 Beijing	**1**	**2**	**P.K. retained**	W.B.: Reads IoT + BIM as living DT. P.K.: Found no evidence of automatic real-time BIM update in Ji et al [29]. The criterion requires a confirmed automatic update.
X3	C3 Taipei 101	**2**	**1**	**P.K. retained**	W.B.: FDS 7.9.1 = partial (design stage). P.K.: The ESP algorithm operates dynamically during the simulated incident. Near-real-time adaptation criterion met.
X5	C3 Taipei 101	**2**	**1**	**P.K. retained**	W.B.: Simulation-only = partial. P.K.: Fully automated sign control in simulation satisfies the criterion; deployment scale does not disqualify the score.
X5	C4 Olympic Venues	**0**	**1**	**P.K. retained**	W.B.: Mobile app = partial actuation. P.K.: Criterion explicitly defines app-only output as absent (score 0). No physical sign update occurred.
Y1	C1 Lille	**1**	**0**	**P.K. retained**	W.B.: Design stage only = 0. P.K.: Pre-computed scenario matching during incident = operator decision support = score 1. Definition of score 1 met.
Y1	C4 Olympic Venues	**2**	**1**	**P.K. retained**	W.B.: App requires user follow-through = partial. PK: Autonomous routing algorithm; mobile app is only the output channel. The full autonomy criterion was met.

Table A5. *Cont.*

Crit.	Case	P.K.	W.B.	Resolution	Interpretation Note and Basis for Retaining the P.K. Score
Y3	C2 Beijing	**1**	**2**	**P.K. retained**	W.B.: Continuous sensor feedback = full closed-loop. P.K.: No evidence of outcome-based model update in Ji et al. [29]; the criterion requires model-mediated feedback.
Y3	C6 IDES Tunnel	**0**	**1**	**P.K. retained**	W.B.: Reversion mechanism = partial feedback. P.K.: Threshold response; not model-mediated. The criterion requires a model-based feedback loop.
Y5	C4 Olympic Venues	**1**	**2**	**P.K. retained**	W.B.: Full autonomy for DT routing. P.K.: No physical signage; the app requires user compliance. The criterion requires an independent sign update.
Y5	C6 IDES Tunnel	**2**	**1**	**P.K. retained**	W.B.: Threshold logic = partial. P.K.: Automated sign change without operator confirmation; operator supervisory role only. Full autonomy criterion met.

Note: Background colours are used only to visually distinguish table sections and improve readability; they do not indicate additional quantitative categories unless explicitly stated.

References

1. Eastman, C.; Teicholz, P.; Sacks, R.; Liston, K. *BIM Handbook: A Guide to Building Information Modeling for Owners, Managers, Designers, Engineers and Contractors*, 2nd ed.; Wiley: Hoboken, NJ, USA, 2011; ISBN 978-0-470-54137-1.
2. Volk, R.; Stengel, J.; Schultmann, F. Building Information Modeling (BIM) for Existing Buildings—Literature Review and Future Needs. *Autom. Constr.* **2014**, *38*, 109–127. [CrossRef]
3. Tang, S.; Shelden, D.R.; Eastman, C.M.; Pishdad-Bozorgi, P.; Gao, X. A Review of Building Information Modeling (BIM) and the Internet of Things (IoT) Devices Integration. *Autom. Constr.* **2019**, *101*, 127–139. [CrossRef]
4. AlQahtani, A.A.S.; Bafail, A.O.; Alothman, A. From Inception to Innovation: A Comprehensive Review and Bibliometric Analysis of IoT-Enabled Fire Safety Systems. *Safety* **2025**, *11*, 41. [CrossRef]
5. Nguyen, V.-Q.; Tran, Q.-B.; Nguyen, V.-H.; Le, X.-H. A Smart Evacuation Guidance System for Large Buildings. *Electronics* **2022**, *11*, 2938. [CrossRef]
6. Khajavi, S.H.; Motlagh, N.H.; Jaribion, A.; Werner, L.C.; Komssi, M.; Holmström, J. Digital Twin for Safety and Security: Perspectives on Building Lifecycle. *IEEE Access* **2023**, *11*, 52339–52356. [CrossRef]
7. Almatared, M.; Liu, H.; Abudayyeh, O.; Hakim, O.; Sulaiman, M. Digital-Twin-Based Fire Safety Management Framework for Smart Buildings. *Buildings* **2024**, *14*, 4. [CrossRef]
8. Lin, J.R. Digital Twin of Buildings and Occupants for Emergency Evacuation: Framework, Technologies, Applications and Trends. *Adv. Eng. Inform.* **2025**, *66*, 103419. [CrossRef]
9. Zhang, X.; Jiang, Y.; Wu, X.; Nan, Z.; Jiang, Y.; Shi, J.; Zhang, Y.; Huang, X.; Huang, G.G.Q. AIoT-Enabled Digital Twin System for Smart Tunnel Fire Safety Management. *Dev. Built Environ.* **2024**, *18*, 100381. [CrossRef]
10. Wang, Q.; Miao, H.; Xu, C.; Hou, B. Comparison of Crowd Evacuation Efficiency under Static Signage and Dynamic Personnel Guidance from an Emotional Contagion Perspective. *Phys. A Stat. Mech. Its Appl.* **2025**, *678*, 130977. [CrossRef]
11. Spinardi, G. Fire Safety Regulation: Prescription, Performance, and Professionalism. *Fire Saf. J.* **2016**, *80*, 64–70. [CrossRef]
12. Meacham, B.J. Performance-Based Design, Expertise Asymmetry, and Professionalism in Fire Safety Regulation. *Regul. Gov.* **2019**, *13*, 145–163. [CrossRef]
13. Meacham, B.J.; van Hees, P. Identifying the Attributes of a Profession in the Practice and Regulation of Fire Safety. *Fire Saf. J.* **2021**, *119*, 103236. [CrossRef]
14. Chow, W.K.; Hung, W.Y.; Lui, G.C.H.; Wan, K.W. Collapse and Imposition in the Regulation of Fire Safety. *Saf. Sci.* **2025**, *181*, 106674. [CrossRef]
15. Çelik, A.; Demirci, Ş.; Polat, O. Simulation-Based Analysis of Evacuation Performance for Assembly Buildings in Accordance with Prescriptive Design. *J. Fac. Eng. Archit. Gazi Univ.* **2024**, *39*, 2343–2352. [CrossRef]

16. Węgrzyński, W.; Sulik, P. The Philosophy of Fire Safety Engineering in the Shaping of Civil Engineering Development. *Bull. Pol. Acad. Sci. Tech. Sci.* **2016**, *64*, 719–730. [CrossRef]
17. Wang, Q.; Li, Y.; Lu, S.; Xin, Y. A Historical Review of Fire Engineering Practice and Advances in China. *Fire Technol.* **2024**, *60*, 711–755. [CrossRef]
18. Abualdenien, J.; Schneider-Marin, P.; Zahedi, A.; Harter, H.; Exner, H.; Meschke, G.; König, M.; Lang, W.; Petzold, F.; Geyer, P.; et al. The Digital Transformation in Fire Safety Engineering Over the Past Decade Through Building Information Modeling. *Fire Technol.* **2022**, *58*, 3317–3351. [CrossRef]
19. Yıldız, M.A. Machine Learning for Fire Safety in the Built Environment: A Bibliometric Insight into Research Trends and Key Methods. *Buildings* **2025**, *15*, 2465. [CrossRef]
20. Torgerson, C. *Systematic Reviews*; Bloomsbury Academic: London, UK, 2003.
21. Nasery, M.M.; Ahmad, S.; Omidvar, B. Proactive Building Fire Risk Management: A Conceptual Framework. *Int. J. Disaster Resil. Built Environ.* **2025**, *17*, 1–17. [CrossRef]
22. Wang, Y.; Li, J.; Chen, Z. Construction and Application of a Fire Safety Resilience Evaluation Model for Road Tunnels Based on the PSR Conceptual Framework. *Results Eng.* **2026**, *30*, 110552. [CrossRef]
23. Mankins, J.C. *Technology Readiness Levels: A White Paper*; Advanced Concepts Office, Office of Space Access and Technology, NASA: Washington, DC, USA, 1995; edited 22 December 2004.
24. Héder, M. From NASA to EU: The Evolution of the TRL Scale in Public Sector Innovation. *Innov. J.* **2017**, *22*, 3.
25. Sauser, B.; Ramirez-Marquez, J.; Verma, D.; Gove, R. From TRL to SRL: The Concept of Systems Readiness Levels. In Proceedings of the Conference on Systems Engineering Research (CSER), Los Angeles, CA, USA, 7–8 April 2006; p. 126.
26. Zhuang, C.; Liu, J.; Xiong, H. Digital Twin-Based Smart Production Management and Control Framework for the Complex Product Assembly Shop-Floor. *Int. J. Adv. Manuf. Technol.* **2021**, *96*, 1149–1163. [CrossRef]
27. Ronchi, E.; Nilsson, D. Guidance for the Model User on Representing Human Behavior in Egress Models. *Fire Technol.* **2017**, *53*, 649–672. [CrossRef]
28. Liu, Z.; Zhang, A.; Wang, W.; Wang, J. Dynamic Fire Evacuation Guidance Method for Winter Olympic Venues Based on Digital Twin-Driven Model. *J. Tongji Univ. Nat. Sci.* **2020**, *48*, 962–971. [CrossRef]
29. Ji, Y.; Wang, W.; Zheng, M.; Chen, S. Real Time Building Evacuation Modeling with an Improved Cellular Automata Method and Corresponding IoT System Implementation. *Buildings* **2022**, *12*, 718. [CrossRef]
30. Yen, H.-H.; Lin, C.-H. Intelligent Evacuation Sign Control Mechanism in IoT-Enabled Multi-Floor Multi-Exit Buildings. *Sensors* **2024**, *24*, 1115. [CrossRef]
31. Wong, H.Y.; Wang, M.; Zhang, X.; Zhang, Y.; Wong, M.C.; Huang, X. Safe Evacuation Framework with Intelligent Dynamic Exit Sign System and Demonstration in Tunnel Fire. *J. Saf. Sci. Resil.* **2025**, *6*, 100183. [CrossRef]
32. Brzezińska, D.; Bryant, P. Risk Index Method—A Tool for Sustainable, Holistic Building Fire Strategies. *Sustainability* **2020**, *12*, 4469. [CrossRef]
33. Wehbe, R.; Shahrour, I. A BIM-Based Smart System for Fire Evacuation. *Future Internet* **2021**, *13*, 221. [CrossRef]
34. Zhang, C. Review of Structural Fire Hazards, Challenges, and Prevention Strategies. *Fire* **2023**, *6*, 137. [CrossRef]
35. Häring, I.; Schönherr, M.; Keller, S.; Finger, J.; Krauß, C.; Riedel, W. Prescriptive Building Regulations, Safety Objectives, and Residual Risk in Germany. *Fire Technol.* **2023**, *59*, 3203–3230. [CrossRef]
36. Ding, Y.; Zhang, Y.; Huang, X. Intelligent Emergency Digital Twin System for Monitoring Building Fire Evacuation. *J. Build. Eng.* **2023**, *72*, 107416. [CrossRef]
37. Zhang, Z.; Tan, L.; Tiong, R.L.K. Fire Emergency Management of Large Shopping Malls: IoT-Based Evacuee Tracking and Dynamic Evacuation. *Alex. Eng. J.* **2024**, *107*, 652–664. [CrossRef]
38. Xie, W.; Zeng, Y.; Zhang, X.; Wong, H.Y.; Zhang, T.; Wang, Z.; Wu, K. AIoT-Powered Building Digital Twin for Smart Firefighting and Super Real-Time Fire Forecasting. *Adv. Eng. Inform.* **2025**, *66*, 103117. [CrossRef]
39. Gu, J.; Wang, J.; Guo, X.; Liu, G.; Qin, S.; Bi, Z. A Metaverse-Based Teaching Building Evacuation Training System with Deep Reinforcement Learning. *IEEE Trans. Syst. Man. Cybern. Syst.* **2023**, *53*, 2209–2222. [CrossRef]
40. Deng, H.; Ou, Z.; Zhang, G.; Deng, Y.; Tian, M. BIM and Computer Vision-Based Framework for Fire Emergency Evacuation Considering Localisation Systems. *Sensors* **2021**, *21*, 3851. [CrossRef]
41. Kim, Y.-J.; Kim, H.; Ha, B.; Kim, W.-T. Advanced Fire Emergency Management Based on Potential Fire Risk Assessment with Informative Digital Twins. *Autom. Constr.* **2024**, *168*, 105722. [CrossRef]
42. Tan, J.K.N.; Zhang, S.; Law, A.W.-K.; Cheung, S.H. Digital Twin-Enabled Evacuation to Improve Individual and Community Resilience of Buildings. *Dev. Built Environ.* **2025**, *22*, 100672. [CrossRef]
43. Rashidian, M.; Malek, M.R. An IFC-Based Framework for Semantic Integration of BIM and Mobile Crowd Sensing in Real-Time Emergency Response. *Adv. Eng. Inform.* **2026**, *67*, 104500. [CrossRef]
44. Chen, D.; Chen, L.; Yang, Y.; Lu, Q.; Hancock, C.; Lock, R.; Sølvsberg, M. A Knowledge-Driven Approach for Automated Fire Safety Compliance Checking in Operational Buildings. *Knowl.-Based Syst.* **2026**, *336*, 115156. [CrossRef]

45. Konopski, P.; Pilch, R.; Bonenberg, W. Responsive Architecture and Fire Safety: A Comparative Review of Regulatory Regimes in the USA, Asia, and the EU/UK, with Implications for Poland. *Sustainability* **2026**, *18*, 3808. [CrossRef]
46. Tyagi, N.; Singh, J.; Singh, S.; Sehra, S.S. A 3D Model-Based Framework for Real-Time Emergency Evacuation Using GIS and IoT Devices. *ISPRS Int. J. Geo.-Inf.* **2024**, *13*, 445. [CrossRef]
47. Mossberg, A.; McNamee, R.; Nyman, H.; Olander, M. A Review of the Swedish Fire Safety Regulation: From the Industry's Perspective. *Fire Mater.* **2021**, *45*, 827–842. [CrossRef]
48. Kodur, V.; Kumar, P.; Rafi, M.M. Fire Hazard in Buildings: Review, Assessment and Strategies for Improving Fire Safety. *PSU Res. Rev.* **2020**, *4*, 1–23. [CrossRef]
49. Cohen, J. Weighted kappa: Nominal scale agreement. *Psychol. Bull.* **1968**, *70*, 213–220. [CrossRef]
50. Landis, J.R.; Koch, G.G. The measurement of observer agreement. *Biometrics* **1977**, *33*, 159–174. [CrossRef]

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